

Pendulum Modeling and Step Response for TFs

EGR 345

GVSU

Outline

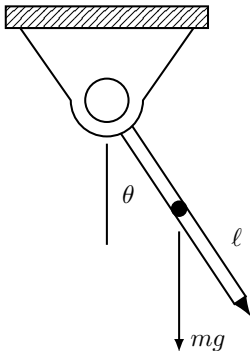
- review: transfer functions
- pendulum modeling
 - we need this for lab
- step function and step response

Review: What is a transfer function?

- how do we find them?
- what can we do with them?

Pendulum Modeling

- How do we find the equation of motion for this system?



Pendulum Modeling 2

- what if there is a force input at the tip of the pendulum?

What is a step function?

- how is $1(t)$ different from just the number 1?

What is a step response?

- and how do we find it?
 - “by hand”
 - with python